

CONTEXTUAL NORMAL RETURN FIELDS FOR POINT-LEVEL LIDAR ANOMALY SEGMENTATION

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ABSTRACT

Point-level LiDAR anomaly segmentation must identify objects outside a normal semantic set while retaining normal objects whose sparse returns may look unfamiliar. We introduce contextual normal return fields to learn the observations supported by the surrounding scene and use these alternatives in the anomaly decision. A context encoder withholds a target angular block and predicts multiple three-dimensional return-rate fields. Each field provides a shared spatial explanation for the block’s rays, and analytic integration converts it into normalized return-distance distributions. A block-level likelihood trains each explanation against a collection of normal returns before marginalizing the hypothesis identity. A point-level decoder then combines the support of the complete hypotheses with actual-observation features to produce one anomaly score. Original nuScenes scans supervise normal prediction, while scans containing auxiliary anomalies and inserted normal objects supervise discrimination. Both insertion roles use the same observation procedure, and grouped supervision gives inserted normal objects an explicit share of the classification objective. The evaluation examines this learning mechanism through source-domain development, controlled component comparisons, and AP, AUROC, and FPR95 on STU.

1 INTRODUCTION

An unfamiliar road object can occupy only a handful of LiDAR returns. A distant pedestrian can be equally sparse and poorly resolved. Point-level anomaly segmentation must distinguish these cases according to their semantic identity while accounting for changes in range, sampling density, and scene layout. STU formalizes this task with point annotations for road objects outside a specified normal semantic set (Nekrasov et al., 2025).

Auxiliary unknown objects provide examples from which a model can learn an anomaly decision. Open-set LiDAR methods obtain such supervision through transformed or inserted objects (Cen et al., 2022; Xu et al., 2025) and connect it to learned anomaly scores (Li et al., 2026c;b). Normal semantic prototypes and reconstructed features provide additional references for interpreting unfamiliar observations (Mosco et al., 2026; Oh et al., 2026). These approaches motivate a complementary learning problem: infer which normal observations the surrounding scene supports, then compare an actual return with those predictions. A context-dependent reference can supply evidence beyond resemblance to the finite set of auxiliary objects used for training.

Constructing this reference requires specifying both the information available to the predictor and the relationship among its outputs. We withhold a target angular block and predict from the measurements in surrounding blocks. The predictor therefore learns to infer the target observations from context. Neighboring LiDAR rays then query a common three-dimensional field, which ties their distance predictions to the same spatial parameters. Several candidate fields represent alternative normal arrangements when the surrounding scene admits more than one continuation.

We formulate these predictions as *contextual normal return fields*. Each hypothesis is a spatial return-rate field whose analytic integral along a ray yields a normalized distribution over return distance. During normal learning, the likelihood of all reliable normal returns in a block is computed under each candidate field before the hypothesis identity is marginalized. This order of operations trains a shared explanation of the block. At inference, a point-level readout retains the likelihood

and prior of each complete hypothesis, allowing the anomaly decision to use both agreement among explanations and differences in their support for the measured return.

The final decision also needs the identity of the observed object. A normal vehicle placed on a road can differ from the structure predicted by its surroundings. Likelihood measures support under a learned distribution, whereas the anomaly label describes semantic membership (Nalisnick et al., 2019; Le Lan & Dinh, 2021). We therefore combine the normal-hypothesis readout with an encoder of the actual scan and learn a single point-level score. Normal prediction and discriminative supervision jointly train this mechanism, following the broader use of complementary normal and anomaly objectives (Grcić et al., 2022; Zavrtanik et al., 2021).

The training observations support this distinction directly. Original nuScenes scans provide normal-field targets, and modified versions provide point-level detection labels. Auxiliary anomalies and normal cars, pedestrians, bicycles, and motorcycles undergo the same road-placement and observation procedure. The inserted normal objects retain their normal labels and receive an explicit share of the classification loss. The decoder thus learns normal retention alongside anomaly detection under the changes introduced by insertion.

Our contribution is an end-to-end anomaly segmentation mechanism built around a contextual normal reference: shared spatial fields define the predictions, a block-level likelihood trains their common explanations, and a point-level readout makes those explanations available to discrimination. The experimental design tests the contribution of this mechanism under fixed source supervision, examines normal-object retention and sparse observations, and evaluates transfer from nuScenes to STU using one score for all three official point-level metrics.

2 RELATED WORK

Point-level unknown detection. REAL learns unknown responses alongside known semantic classes, and LiON uses diverse synthetic objects to train point-wise abstention (Cen et al., 2022; Xu et al., 2025). Relative energy learning and neural distribution priors further connect anomaly supervision to learned segmentation scores (Li et al., 2026c;b). LIDO constructs complementary prototype, entropy, and contrastive evidence from normal training data (Mosco et al., 2026). CO-VAL studies visibility interventions for LiDAR anomaly learning (Yang et al., 2026). These methods establish semantic representation, auxiliary anomalies, and observation structure as useful sources of supervision. Our method develops an explicit conditional normal-return representation that a point-level decoder reads at inference.

Normal prediction for anomaly segmentation. Image resynthesis and inpainting provide normal references for road anomaly detection (Lis et al., 2024). DRAEM couples reconstruction with discriminative anomaly segmentation, while DenseHybrid combines normal-density and outlier-discrimination objectives (Zavrtanik et al., 2021; Grcić et al., 2022). InTra excludes the target image patch before predicting it from its neighborhood, and UniAD restricts reconstruction pathways to address input copying (Pirnay & Chai, 2022; You et al., 2022). In three dimensions, DOODLE augments LiDAR anomaly scores with diffusion-based feature reconstruction, and GPFlow models normal feature distributions with Gaussian prototypes (Oh et al., 2026; Li et al., 2026a). We predict distributions of measured distances from shared spatial hypotheses and retain the support of each hypothesis in the anomaly readout.

Conditional prediction and geometric representation learning. Blind-spot denoising predicts a distribution from surrounding observations and combines it with the measured target (Laine et al., 2019; Krull et al., 2020). Conditional neural processes learn context-dependent prediction rules with uncertainty over queried outputs (Garnelo et al., 2018). In point clouds, masked reconstruction learns geometric representations (Pang et al., 2022); UniPAD and Ponder connect three-dimensional fields to rendered observations (Yang et al., 2024; Huang et al., 2023). Probabilistic LiDAR fields also connect ray attenuation with return distributions (McDermott & Rife, 2024). Our formulation brings shared field prediction into the anomaly decision itself through a learned comparison with observed points.

Source-domain geometry and data construction. Scene completion, sensor-aware augmentation, and density variation support cross-domain LiDAR perception (Yi et al., 2021; Ryu et al., 2023; Kim et al., 2023). Mix3D and PolarMix vary the surroundings of real point-cloud instances while preserving their labels (Nekrasov et al., 2021; Xiao et al., 2022). We use source observations to construct both auxiliary anomalies and normal insertion controls, and organize the learning problem around their point-level semantic labels. This connects the training construction directly to the normal-reference mechanism.

3 CONTEXTUAL NORMAL RETURN FIELDS

3.1 POINT-LEVEL PREDICTION FROM OBSERVATIONS AND NORMAL CONTEXT

Let $X = \{(x_i, I_i)\}_{i=1}^n$ denote one LiDAR scan, where $x_i \in \mathbb{R}^3$ is a return coordinate and I_i its intensity. Each return has a direction d_i and distance r_i relative to the scan reference origin. The network produces a scalar $a_i \in \mathbb{R}$ whose ordering represents anomaly evidence. Training labels assign reliable normal points to zero, auxiliary anomaly points to one, and ambiguous points to an ignored set.

The model contains an actual-observation encoder and a normal-context predictor with separate parameters. The observation encoder uses LitePT-S (Yue et al., 2026) with a point-detail pathway to obtain $f_i \in \mathbb{R}^{64}$. It integrates multi-scale scene features, raw-point detail, and the point’s position within its voxel. The normal predictor partitions directions into angular blocks at three scales. For each target block B , it reads the surrounding blocks C_B and predicts a distributional reference for the target rays. The two pathways meet in the point-level readout shown in Figure 1.

The normal predictor uses a separate pointwise encoder and pools its features independently within each angular block. For a target block, the hypothesis decoder reads the other 24 block representations in its 5×5 neighborhood. Learned hypothesis queries attend to these representations together with the fixed block location and scale. Target measurements are withheld before any cross-block interaction. At each scale, the predicted field parameters serve all target rays in that block.

3.2 SHARED SPATIAL HYPOTHESES AND RETURN DISTRIBUTIONS

For a target block B , the context decoder produces K normal hypotheses with weights π_{Bk} . Each hypothesis is a positive spatial rate field formed by M anisotropic Gaussian components:

$$\pi_{Bk} = \text{softmax}(b_B)_k, \quad (1)$$

$$\lambda_{Bk}(x) = \sum_{m=1}^M a_{Bkm} \exp \left[-\frac{1}{2} (x - c_{Bkm})^\top \Sigma_{Bkm}^{-1} (x - c_{Bkm}) \right]. \quad (2)$$

The network predicts the centers c_{Bkm} , positive amplitudes a_{Bkm} , and full positive-definite covariances Σ_{Bkm} . The rate has units of inverse distance. Components within a hypothesis jointly describe one spatial arrangement, and the mixture represents alternative arrangements supported by the context.

A ray queries this representation through $x_i(r) = o_i + r d_i$, with unit direction d_i . We model the distance of a recorded return within the interval $[L, U]$. Define the accumulated rate and its normalization by

$$H_{ik}(r) = \int_L^r \lambda_{Bk}(o_i + t d_i) dt, \quad Z_{ik} = 1 - \exp[-H_{ik}(U)]. \quad (3)$$

The conditional return-distance density is

$$p_{ik}(r | C_B, d_i, \text{recorded return}) = \frac{\lambda_{Bk}(o_i + r d_i) \exp[-H_{ik}(r)]}{Z_{ik}}, \quad L \leq r \leq U. \quad (4)$$

The exponential term couples the rate at distance r to the accumulated rate at shorter distances. Normalization conditions the prediction on a recorded return in the interval. All target rays in B use the same λ_{Bk} , so a change to a spatial component affects several distance predictions through their ray geometry.

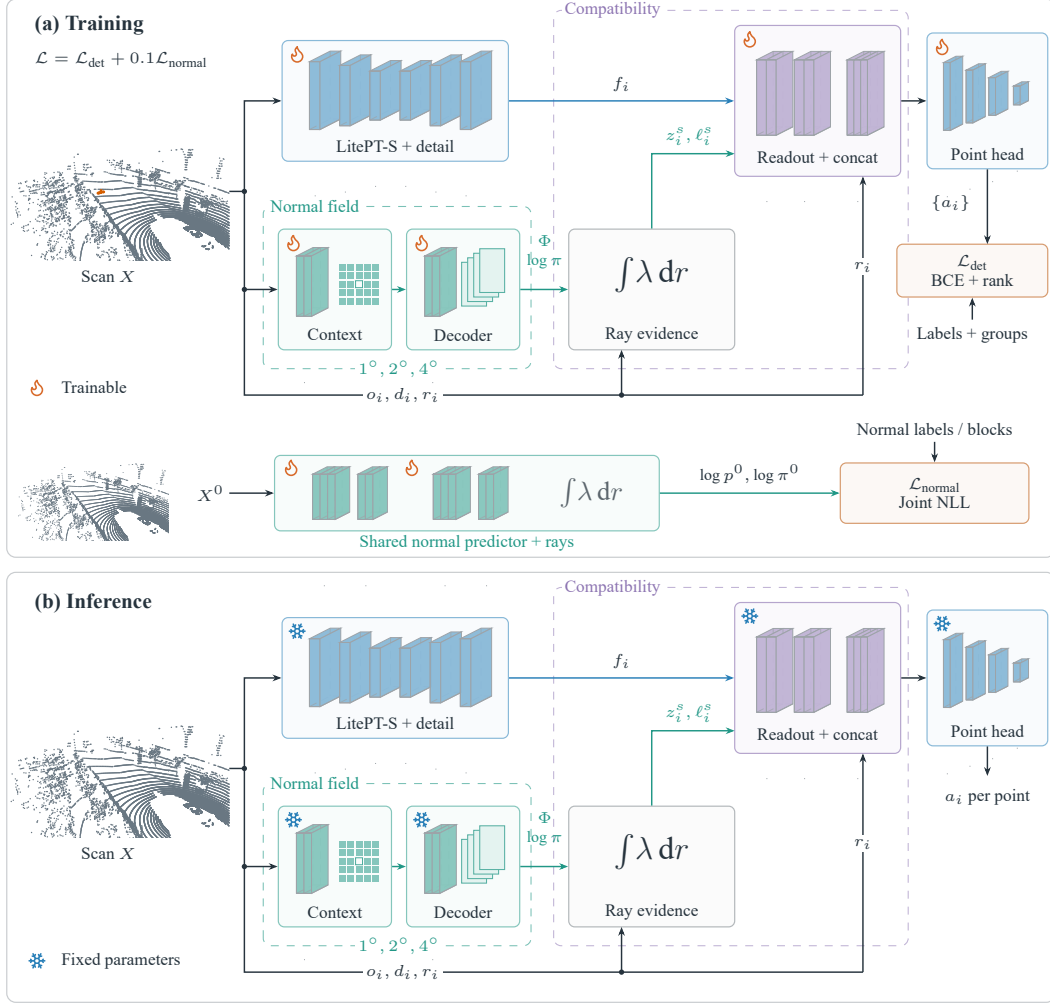


Figure 1: Contextual normal return fields. (a) Detection and original-scan likelihood jointly train the model. (b) Inference outputs one anomaly score per point. Ray evidence computes hypothesis log densities and marginal log density ℓ_i^s , bundling densities with decoder log priors as z_i^s . Readout includes query formation, hypothesis attention and feature concatenation. The auxiliary path shares the normal predictor. Arrows show forward data dependencies; dashed boxes group computations. Flames mark trainable modules; snowflakes mark fixed inference parameters. Gray operators are analytic. Layer glyphs are schematic. The real nuScenes scenes share a viewpoint; orange marks auxiliary labels.

Restricting a three-dimensional Gaussian to a ray gives a one-dimensional Gaussian response. Consequently, H_{ik} is evaluated analytically through Gaussian cumulative probabilities. This provides differentiable distance densities without discretizing each ray into depth samples. Appendix A gives the derivation.

3.3 LEARNING A SHARED NORMAL EXPLANATION

Let $\mathcal{I}_B$ index the reliably labeled normal returns within $[L, U]$ in block B of an original scan. Write their distances as $R_B = \{r_i\}_{i \in \mathcal{I}_B}$ and directions as $Q_B = \{d_i\}_{i \in \mathcal{I}_B}$. The normal-learning objective assigns a common hypothesis to these returns:

$$P(R_B | C_B, Q_B, N) = \sum_{k=1}^K \pi_{Bk} \prod_{i \in \mathcal{I}_B} p_{ik}(r_i), \quad (5)$$

where N denotes normal supervision. Each candidate field is evaluated against the same normal returns before the hypothesis identity is marginalized. The block loss is

$$\mathcal{L}_B = -\frac{1}{|\mathcal{I}_B|} \log \sum_{k=1}^K \pi_{Bk} \exp \left[\sum_{i \in \mathcal{I}_B} \log p_{ik}(r_i) \right]. \quad (6)$$

We average this loss over blocks containing normal targets, then over scales and reference scans. Each block contributes its joint negative log-likelihood normalized by its number of normal returns, and the outer average weights populated blocks equally. A shared hypothesis couples the learning signals from their rays through one spatial field. Section 5 specifies a comparison with independent per-point hypothesis mixing.

3.4 HYPOTHESIS COMPATIBILITY WITH THE OBSERVED POINT

The point-level decoder compares each actual return with the complete hypotheses predicted for its block. For scale s , a shared map encodes the hypothesis log density at the observed distance and its context-derived log prior:

$$v_{ik}^s = \psi[\log p_{ik}^s(r_i), \log \pi_{Bk}^s] + \delta_s, \quad q_i = W_q f_i + \rho(r_i/U). \quad (7)$$

Here δ_s is a learned scale embedding, and the query q_i combines observed point features with measured distance. Each density is computed from all Gaussian components of its spatial hypothesis through Equation 4. The decoder attends to the resulting hypothesis representations:

$$e_i^s = \text{Attn}_k(q_i, \text{key}(v_{ik}^s), \text{value}(v_{ik}^s)). \quad (8)$$

The 64-dimensional hypothesis representation is split into 32-dimensional keys and values. Four attention heads aggregate evidence across the K hypotheses. The query selects useful evidence for the point-level decision, while the field parameters and their priors are predicted from the surrounding blocks.

We additionally provide the point’s marginal normal log density,

$$\ell_i^s = \log \sum_k \pi_{Bk}^s p_{ik}^s(r_i), \quad a_i = g[f_i, e_i^1, e_i^2, e_i^3, \ell_i^1, \ell_i^2, \ell_i^3]. \quad (9)$$

The absolute density complements the attention-weighted features when all hypotheses assign little support to the return. The output a_i is the point-level anomaly score. The normal objective in Equation 5 shapes the shared representation during training; Equation 9 specifies its point-wise use during inference.

3.5 JOINT OPTIMIZATION

Detection supervision and normal prediction train the full model together. Within each effective batch, we average binary cross-entropy separately over anomaly points, inserted normal points, and other reliable normal points. Their respective weights are $1/2$, $1/4$, and $1/4$. Group averaging preserves the contribution of inserted normal objects even when background points dominate the scan. With this classification objective $\mathcal{L}_{\text{BCE}}$, we optimize

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha(t)(\tilde{\mathcal{L}}_{\text{AP}} + 0.1\tilde{\mathcal{L}}_{\text{AUC}} + 0.1\tilde{\mathcal{L}}_{\text{FPR95}}) + 0.1\mathcal{L}_{\text{normal}}. \quad (10)$$

The ranking terms operate on the same a_i across an effective batch of scans. They address the ordering of anomaly points, overall positive-negative separation, and the normal tail at high anomaly recall. Their weight $\alpha(t)$ increases from zero to one between 10% and 20% of the update budget. Appendix B defines the surrogates and their sampling. Detection gradients pass through the compatibility decoder and the return densities into the normal predictor, while original normal scans provide the likelihood supervision.

4 SOURCE TRAINING OBSERVATIONS

The two learning objectives use paired views of the same source scene. An original scan supplies reliable normal returns for field prediction. An observation containing an inserted object supplies the point labels for discrimination. Auxiliary anomalies and normal objects share the insertion procedure, so the training labels teach the decoder which inserted objects should be retained as normal.

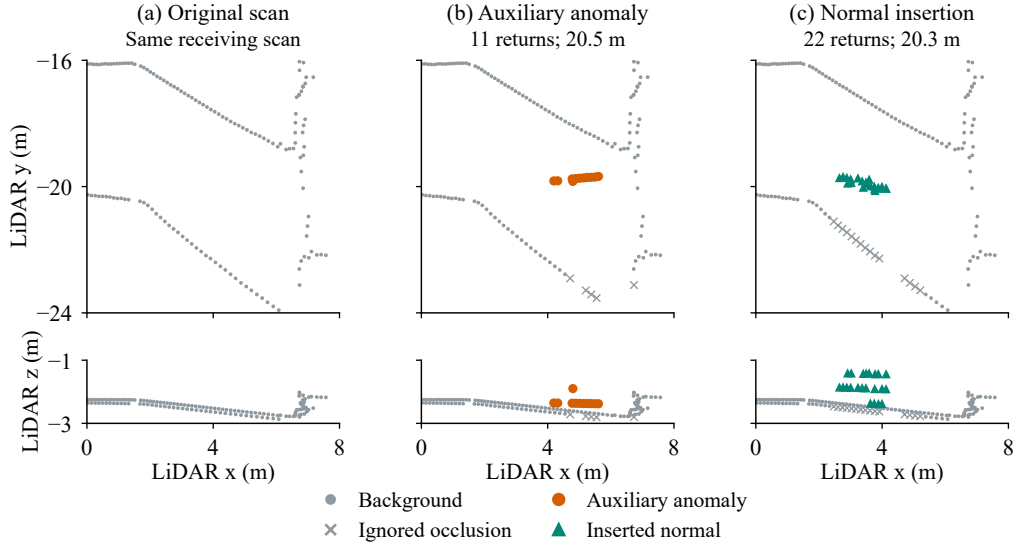


Figure 2: Realized source observations from one nuScenes training scan. Columns show the original, an auxiliary toolbox insertion (11 returns), and a normal motorcycle insertion (22 returns). Top and bottom rows show horizontal and height projections of the same crop, with equal distance scales on both axes. The insertions use independent world poses and the same observation procedure. Crosses mark background returns retained with ignored supervision in uncertain occlusion regions.

Backgrounds and semantic supervision. We construct training and development observations from the official nuScenes training and validation partitions (Caesar et al., 2020). They contain 700 and 150 scenes, respectively. A source point receives normal supervision when its annotated meaning agrees with the target normal set. Ambiguous source categories receive ignored labels. Camera observations support donor verification and local label refinement; model inputs are LiDAR coordinates and intensity.

Auxiliary anomalies and normal controls. The auxiliary object collection contains 40 verified instances, divided into 27 training and 13 development donors. They include bags, refuse containers, carts, toolboxes, and suitcases. These nuScenes objects supply auxiliary anomaly supervision for evaluation on STU’s separately defined target anomaly set (Nekrasov et al., 2025). Normal donors are adult pedestrians, bicycles, motorcycles, and cars. Auxiliary donors receive anomaly supervision after insertion, while inserted normal objects retain normal supervision.

Placement and recorded observations. Each placement fixes one original-size donor at a world pose supported by measured road points. Independent placements cover an unrestricted base placement and requested observation ranges of 20–30, 30–40, and 40–50 m. Vehicle motion then determines the sequence of viewing distances and directions. For each receiving scan, its recorded ray directions intersect the donor’s measured open surface. A closer supported intersection replaces the corresponding return. Observations are retained within the views supported by the measured donor surface. Background points in uncertain occlusion regions retain their coordinates with ignored supervision.

Paired learning records. Every modified observation is linked to its unmodified scan. The normal-field targets always come from this original, including when the inserted object is normal. All versions of a scene remain in its original partition, and donor instance identities are partitioned with their source scenes. Table 1 summarizes the resulting records. Each original background scan appears once in the reading index.

Table 1: nuScenes source data. Modified records share their original scene and scan identities; the first two rows describe the underlying background observations.

	Training	Development
Background scenes	700	150
Original scans	28,130	6,019
Auxiliary anomaly donors	27	13
Anomaly insertion records	23,795	4,699
Normal insertion records	20,895	4,861
Total reading records	72,820	15,579

Table 2: Point-level anomaly segmentation on STU. Values are percentages. Higher AP and AUROC and lower FPR95 are better. Each metric triplet uses one checkpoint and the same point scores.

Method	Validation			Hidden test		
	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$
NDP (Li et al., 2026b)						
REL (Li et al., 2026c)						
LIDO (Mosco et al., 2026)						
COVAL (Yang et al., 2026)						
Ours						

5 EXPERIMENTS

5.1 EVALUATION AND IMPLEMENTATION

Evaluation. The target benchmark is STU, with its public 19-sequence anomaly validation partition and 51-sequence hidden anomaly test partition (Nekrasov et al., 2025). The point-level evaluation uses valid labels in the 2.5–50 m interval and aggregates eligible scans containing at least five anomaly points. We report average precision (AP), area under the receiver operating characteristic curve (AUROC), and the false-positive rate at 95% anomaly recall (FPR95). All three metrics use the same checkpoint and unnormalized point score. Source development uses the nuScenes partition described in Section 4. Normal insertion observations additionally support an analysis of normal-point scores and false alarms.

Training. The observation backbone is initialized with official nuScenes LitePT-S weights; all added modules are initialized from scratch. All parameters are optimized jointly using AdamW, with peak learning rates of 2×10^{-4} for the backbone and 2×10^{-3} for the added modules. The training protocol specifies 2,500 updates with eight scans per effective batch, corresponding to 20,000 scan visits. Full source-development evaluation is scheduled at updates 1,250 and 2,500. Source AP selects one checkpoint, with endpoint results reported separately in Table 7. STU observations are excluded from optimization and checkpoint selection in this source-trained protocol. The model uses 5 cm voxels, angular scales of 1° , 2° , and 4° , four normal hypotheses, and eight Gaussian components per hypothesis. Computation uses FP32. Appendix C supplies the remaining details.

5.2 BENCHMARK COMPARISON

The target comparison in Table 2 evaluates the complete model against LiDAR anomaly methods on the public validation and hidden-test partitions separately. AP measures precision across anomaly recall levels, AUROC summarizes positive-negative ordering, and FPR95 focuses on normal-point rejection at high anomaly recall. Their joint reporting characterizes detection and normal retention using one score. Appendix D specifies the training resources for each compared model.

Table 3: Controlled component comparison on STU validation. Each row specifies a separately trained variant under the common source training setup.

Variant	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$
Direct point classifier			
Point-wise mixture objective			
Marginal-density readout			
Full model			

Normal-control scores and range/point-count analysis

Figure 3: Normal retention and sparse anomaly observations: score distributions for inserted normal objects, range-stratified measurements, and measurements grouped by observed object point count.

5.3 CONTRIBUTION OF THE NORMAL REPRESENTATION

The component study uses a common source dataset, backbone initialization, optimization budget, and detection objective. A direct point classifier supplies the observation-only reference. A point-wise mixture variant replaces the block objective in Equation 5 with $\prod_i \sum_k \pi_k p_{ik}(r_i)$ while retaining the same field architecture and inference decoder. A marginal-density readout retains the observation feature f_i and the three log densities ℓ_i^s in the final head. The complete model additionally reads the individual hypotheses through Equations 7–8. These comparisons separate the contribution of a normal reference, shared-hypothesis training, and hypothesis-specific evidence.

5.4 NORMAL CONTROLS AND SPARSE OBSERVATIONS

Normal insertion controls test whether the learned decision retains normal objects after road placement and changes in observation. Their score distributions in Figure 3 complement anomaly measurements grouped by range and observed object point count. Range changes sampling density, while point count describes the evidence actually available for each object. Donor-wise and object-type summaries further show how individual source objects contribute to pooled performance. These analyses connect the point-level metrics to the observations on which the model succeeds or fails.

5.5 QUALITATIVE ANALYSIS

The qualitative comparison pairs point-level predictions with the normal-return distributions used by the decoder (Figure 4). Inputs, annotations, direct-classifier outputs, and full-model scores expose detection errors and retained normal objects in the same scenes. Selected rays relate the measured return to the support of the predicted hypotheses. Together, these views examine how the normal reference contributes to decisions at object boundaries and under sparse observations.



Point-level predictions and normal-return distributions

Figure 4: Qualitative comparison on STU validation: input points, annotations, direct-classifier predictions, full-model scores, and normal-return distributions for selected rays.

6 CONCLUSION

We introduced contextual normal return fields for point-level LiDAR anomaly segmentation. Surrounding observations predict shared spatial hypotheses, analytic ray integration converts them into distance distributions, and a learned decoder combines their support with actual-observation features. Block-level normal learning and point-level discrimination train this mechanism jointly. Original scans provide normal targets, while auxiliary anomalies and normal insertion controls teach the final semantic decision. The resulting formulation connects context-based normal prediction directly to point-level anomaly segmentation.

AI USE STATEMENT

We used OpenAI Codex to assist with literature search and synthesis, refinement of research hypotheses and the model formulation, mathematical derivations, training-data construction and annotation review, experimental design, implementation and debugging, and interpretation of development results. Codex also assisted with manuscript drafting and editing, reference organization, figure design and plotting code, and document formatting. Synthetic point clouds are produced by the nuScenes-based geometric procedure in Section 4; AI assistance included designing and implementing that procedure. Checks of the assisted implementation include numerical evaluation of return densities and gradients, metric recomputation, and verification of source labels and generated observations. The authors take responsibility for the final text, claims, code, and data artifacts.

REPRODUCIBILITY STATEMENT

Section 3 defines the model and learning objectives, and Section 4 describes the source-data construction and partitioning. Appendices A–C provide the analytic derivation, loss definitions, model dimensions, optimization settings, and sampling procedure. Appendix D records the training resources and observation coverage.

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A ANALYTIC GAUSSIAN INTEGRATION ALONG A RAY

Consider one component with covariance $\Sigma = AA^\top$ and rate

$$\lambda_m(x) = a \exp\left[-\frac{1}{2}(x - c)^\top \Sigma^{-1}(x - c)\right]. \quad (11)$$

For the ray $x(r) = o + rd$, define $v = A^{-1}d$ and $w = A^{-1}(o - c)$. Completing the square gives

$$\mu = -\frac{v^\top w}{v^\top v}, \quad \tau = (v^\top v)^{-1/2}, \quad \log h = \log a - \frac{1}{2}\|w + \mu v\|^2, \quad (12)$$

$$\lambda_m(o + rd) = h \exp\left[-\frac{(r - \mu)^2}{2\tau^2}\right]. \quad (13)$$

The integrated component rate is therefore

$$\int_L^b \lambda_m(o + rd) dr = h\tau\sqrt{2\pi} \left[\Phi\left(\frac{b - \mu}{\tau}\right) - \Phi\left(\frac{L - \mu}{\tau}\right) \right], \quad (14)$$

where Φ is the standard Gaussian cumulative distribution function. Summing these terms over m gives $H_{ik}(b)$. The conditional density integrates to one because

$$\int_L^U \lambda_{Bk}(o + rd) e^{-H_{ik}(r)} dr = \left[-e^{-H_{ik}(r)} \right]_L^U = 1 - e^{-H_{ik}(U)}. \quad (15)$$

All field and density operations are differentiable with respect to the predicted parameters. We compute interval probabilities in the log domain, using stable tail and narrow-interval expressions. Point queries are processed in chunks of 2,048.

The implemented observation model uses the scan reference origin and conditions on a recorded return in $[2.5, 50]$ m. Its learned rate is a representation for normal distance prediction. The conditioning event specifies which recorded measurements enter the likelihood objective. For returns outside $[L, U]$, log-density inputs to the readout are set to zero; these returns remain available as context.

B POINT-LEVEL TRAINING OBJECTIVES

Let $\mathcal{P}$ and $\mathcal{N}$ be anomaly and normal points across one effective batch. Partition $\mathcal{N}$ into inserted normal controls $\mathcal{C}$ and other reliable normal points $\mathcal{B}$. With all three groups present, classification supervision is

$$\mathcal{L}_{\text{BCE}} = \frac{1}{2|\mathcal{P}|} \sum_{p \in \mathcal{P}} \log(1 + e^{-a_p}) + \frac{1}{4|\mathcal{C}|} \sum_{c \in \mathcal{C}} \log(1 + e^{a_c}) + \frac{1}{4|\mathcal{B}|} \sum_{b \in \mathcal{B}} \log(1 + e^{a_b}). \quad (16)$$

For an absent group, its term is omitted and the remaining group weights are normalized to sum to one. The source batch construction supplies all three groups. Insertion identity defines the supervision groups and is excluded from the model input. Ignored points remain available as context and are excluded from these supervised sets.

Ranking samples. We uniformly sample up to 256 anomaly anchors into $\mathcal{A} \subseteq \mathcal{P}$. All anomaly scores remain available when computing positive ranks and the soft recall threshold. Up to 4,096 normal points form $\mathcal{S}_-$. When $|\mathcal{N}| > 4,096$, this set contains the 512 highest-scoring normal points and 3,584 uniformly sampled points from the remainder. The former have weight one and the latter have weight $(|\mathcal{N}| - 512)/3584$. Otherwise all normal points are retained with unit weights. Denote these weights by w_n .

Average-precision surrogate. With temperature $\tau = 1$ and logistic function σ , the soft positive and negative ranks of anomaly anchor p are

$$R_p^+ = 1 + \sum_{q \in \mathcal{P} \setminus \{p\}} \sigma\left(\frac{a_q - a_p}{\tau}\right), \quad R_p^- = \sum_{n \in \mathcal{S}_-} w_n \sigma\left(\frac{a_n - a_p}{\tau}\right), \quad (17)$$

$$\tilde{\mathcal{L}}_{\text{AP}} = 1 - \frac{1}{|\mathcal{A}|} \sum_{p \in \mathcal{A}} \frac{R_p^+}{R_p^+ + R_p^-}. \quad (18)$$

Pairwise separation. The AUROC-oriented term uses a smooth positive-negative ordering loss:

$$\tilde{\mathcal{L}}_{\text{AUC}} = \frac{1}{|\mathcal{A}| |\mathcal{N}|} \sum_{p \in \mathcal{A}} \sum_{n \in \mathcal{S}_-} w_n \text{softplus} \left(\frac{a_n - a_p}{\tau} \right). \quad (19)$$

Normal tail at high recall. The soft threshold θ satisfies

$$\frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \sigma \left(\frac{a_p - \theta}{\tau} \right) = 0.95. \quad (20)$$

We differentiate this threshold implicitly and optimize the weighted normal exceedance:

$$\tilde{\mathcal{L}}_{\text{FPR95}} = \frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{S}_-} w_n \sigma \left(\frac{a_n - \theta}{\tau} \right). \quad (21)$$

For $u_p = (a_p - \theta)/\tau$, the threshold derivative is $\partial\theta/\partial a_p = \sigma'(u_p)/\sum_q \sigma'(u_q)$. Ranking terms are zero when either class is absent. The BCE term continues to supervise the present groups. The schedule is

$$\alpha(t) = \text{clip} \left(\frac{t/T - 0.1}{0.1}, 0, 1 \right), \quad T = 2500. \quad (22)$$

C ARCHITECTURE AND OPTIMIZATION DETAILS

The model contains 13,208,930 trainable parameters. Table 4 lists its dimensions. The 2° and 4° grids are nested coarsenings of the 1° grid. A target block reads the 24 other blocks in its 5×5 neighborhood at the selected scale. Empty neighbors use an attention mask; an entirely empty neighborhood uses a learned context token. The context predictor uses LayerNorm and block-local pooling.

Table 4: Implemented model dimensions. The hypothesis map uses the same parameters across hypotheses and angular scales.

Component	Configuration
Observation backbone	LitePT-S; 5 cm voxel size
Actual point state	64-dimensional scene, detail, and within-voxel terms summed
Context point encoder	$[x/U, y/U, z/U, I, d_x, d_y, d_z]$; $7 \rightarrow 64 \rightarrow 64$
Block representation	Mean/max pooling; 128 dimensions
Hypothesis decoder	Four queries; two four-head cross-attention layers; feed-forward width 256
Spatial fields	Four hypotheses, each with eight full-covariance Gaussian components
Hypothesis map ψ	$2 \rightarrow 16 \rightarrow 64$; 32-dimensional keys and values
Compatibility attention	Four heads; eight dimensions per head
Final point head	$163 \rightarrow 128 \rightarrow 64 \rightarrow 1$

The context and field decoder share parameters across scales. Covariances use Cholesky factors, and positive parameterizations give positive amplitudes. Hypothesis queries are initialized separately. Field centers are initialized across the valid distance interval in a coordinate basis determined by the target angular block. The actual-point query combines the 64-dimensional point state projected to 32 dimensions with a $1 \rightarrow 16 \rightarrow 32$ distance encoding. Hidden layers in the final head use normalization and GELU.

Within each context-attention layer, keys and values are projected once per independent angular block and then gathered for the original 24-neighbor attention. This reuses the same projections while preserving the attention parameters and target-block exclusion.

AdamW uses $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, and weight decay 0.005, with zero decay for biases and normalization parameters. Learning rates warm up for the first 5% of updates from 0.1 of their peak

value, then follow cosine decay to 0.01 of that value. The gradient norm is clipped at one. The fixed source run uses random seed zero and the preconstructed observations directly. Ties in checkpoint AP are broken by lower FPR95 and then higher AUROC.

Each effective batch contains three observations with at least five anomaly points, one with one to four anomaly points, two with observed normal insertions, and two original normal scans. Within each insertion role, measured range bands are visited in proportion to their record counts in that role pool throughout training. Within a range band, ordering interleaves donor instances, receiving scenes, point-count bins, placement versions, and observation segments. Temporally separated frames precede adjacent frames. Original scans are interleaved by scene. Anomaly ranges use the median supervised-point distance; normal controls use the median band of their supervised inserted-point histogram. Records are reused only after a complete traversal of their role pool and never within the same effective batch.

The full scan supplies context to both encoders. During training, the compatibility readout and point head query only supervised points. Original normal references query ray densities only at their reliable normal targets and require only the normal predictor. Scores from all eight scans are collected before computing the joint detection objective. Cached score gradients are then replayed through per-scan forward computations, with the normal-reference loss added to the same parameter update. Normalization state advances once per training observation. This arrangement retains the effective-batch ranking objective while processing the backbone one scan at a time. Inference returns a score for every observed point.

D TRAINING RESOURCES AND OBSERVATION COVERAGE

Table 5 records training resources alongside the target comparison. The model uses official nuScenes backbone initialization and nuScenes auxiliary objects. The designated source run uses nuScenes for optimization, development evaluation, and checkpoint selection. STU validation is reported as a public development partition, and the hidden partition supplies the separate test evaluation.

Table 5: Training and development resources for the checkpoints in the benchmark comparison.

Method	Backbone training sources	Auxiliary anomalies	Target-domain use
NDP			
REL			
LIDO			
COVAL			
Ours	nuScenes	nuScenes donors	Public validation reporting

The source background partitions contain disjoint acquisition logs, and donor instance identities follow the source partition. The 40 auxiliary donors originate from 21 training and 12 development scenes. Their single-observation point counts range from 8 to 601. Normal donors come from adult pedestrians, bicycles, motorcycles, and cars. Original debris and pushable/pullable labels remain ignored; verified donor surfaces receive auxiliary anomaly labels after insertion. Reliable normal supervision includes compatible source labels and 4,752 manually refined building points across six source frames.

Every placement uses the donor at its measured size and a fixed world pose. A support point lies at least one meter inside the convex hull of local measured road points, and collision checks use scene observations. The receiving scan supplies the queried directions. Surface intersections replace recorded slots when they are nearer and supported by donor visibility. All independent placement versions remain with the same original scene. Table 6 gives counts derived from the resulting observations.

Among synthesized auxiliary anomaly returns within the 2.5–50 m evaluation interval, refuse containers contribute 44.7% of training points and 86.8% of development points. Donor-wise and type-wise summaries therefore accompany pooled point-level evaluation, together with range and point-

Table 6: Observed source coverage. Range statistics use the saved returns. The five-point condition follows the point-level evaluation eligibility rule.

	Training	Development
Successful auxiliary placements	1,828	366
Successful normal placements	1,542	359
Scans with at least five anomaly points	10,486	1,796
Scans with one to four anomaly points	10,180	1,985
Auxiliary anomaly points in range	235,996	52,341
At least five anomaly points beyond 30 m	494	223
At least five inserted normal points beyond 30 m	575	129

count strata. The original scans, insertion records, and donor identities define the units for these summaries.

E ADDITIONAL RESULTS

Table 7: Source-development evaluation and computational measurements. Time per scan includes both model pathways for the full model.

Model	AP $\uparrow$	AUROC $\uparrow$	FPR95 $\downarrow$	Time per scan
Direct point classifier				
Full model, source-selected				
Full model, update 2,500				